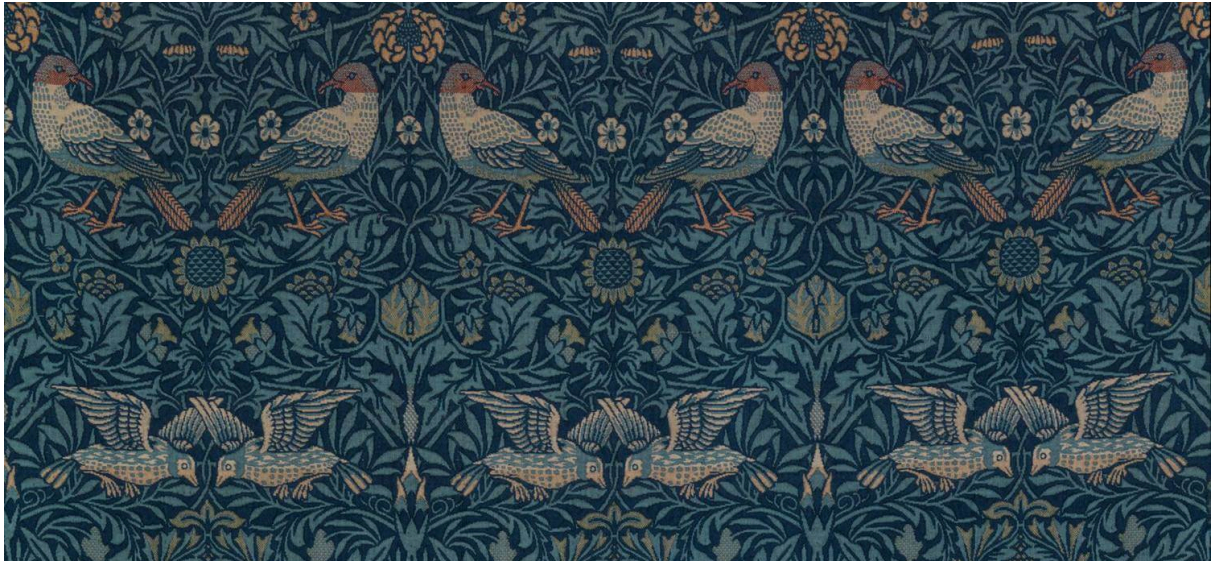




EY Hack

GitHub repo: <https://github.com/sanvviratthore/tata-loan-chatbot>

Agent architecture

Master agent (Orchestrator)

- sales agent (loan offers)
- verification agent (Kyc & identity)
- Credit Bureau " (loan detectⁿ)
- underwriting " (decisions)
- document " (pdf genⁿ)
- debt consolidation agent (hero feature)

Our Hero feature

- ① Customer acquisition cost: Rs. 0
normal: spend Rs. 20k - 5k in ads to acquire new customer
this feature: steal customers from comp^y for free.

- ② Larger loan size = more revenue.
 - Regular loan: Rs. 1-2 lakhs
 - Consolid: " : Rs. 3-5 " (combⁿ multiple loans)
 - more interest income for tata.

Math :-

Before consolid. →

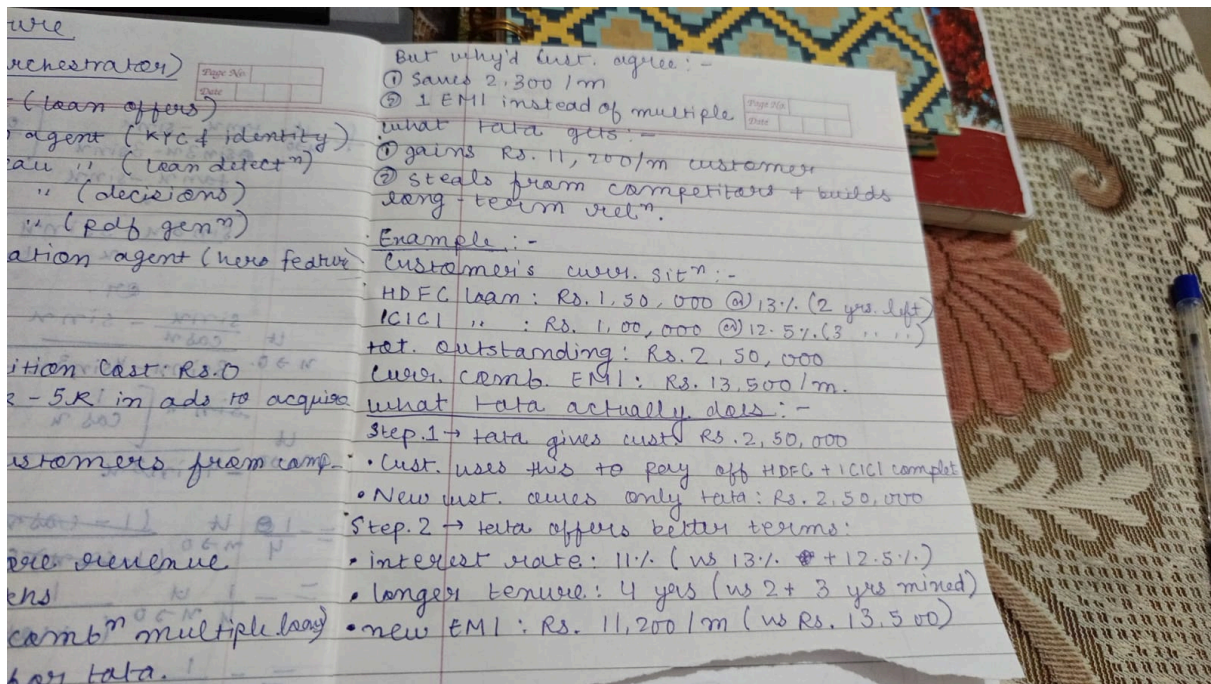
Customer pays: HDFC (Rs. 8k) + ICICI (Rs. 5k) = Rs. 13,500/m.

Tata gets: Rs. 0

After consolid. →

Customer pays tata: Rs. 11,200/m (saves Rs. 2,300)

tata gets Rs. 11,200/m from a cust. who was giving money to competitors.



MAIN FLOW: Successful Auto-Detection & Consolidation

1. Initial Contact & Identity

USER: "I need a personal loan"

MASTER AGENT:

"Welcome to Tata Capital! To find your best loan options, I need to verify your identity.

Please provide your PAN card number or registered mobile number:"

2. Identity Verification & Auto-Scan

USER: "ABCP1234X" (PAN)

VERIFICATION AGENT:

"✅ Identity Verified - Raj Sharma

🔍 Scanning your credit profile with authorized consent..."

CREDIT BUREAU AGENT:

" Credit Profile Analysis Complete!"

Active Loans Found:

- HDFC Personal Loan: ₹1,50,000 (EMI: ₹8,000)
- ICICI Car Loan: ₹1,00,000 (EMI: ₹5,500)
- Total Monthly Outgo: ₹13,500
- Credit Score: 780/900 

3. Smart Consolidation Offer

DEBT CONSOLIDATION AGENT:

" DEBT CONSOLIDATION OPPORTUNITY!"

Your Current Situation:

 Monthly Outgo: ₹13,500 (2 loans)

Tata Smart Consolidation:

 Single EMI: ₹11,200/month (₹3,50,000 @ 11%)

 INSTANT SAVINGS: ₹2,300/month!

Would you like to simplify your finances? 

4. Processing & Approval

USER: "Yes, proceed"

UNDERWRITING AGENT:

" Loan Approved!"

- Amount: ₹3,50,000
- Interest: 11% | Tenure: 4 years
- EMI: ₹11,200/month"

DOCUMENT AGENT:

" Consolidation Sanction Letter Generated!
[Download PDF]"

5. Completion

MASTER AGENT:

"Congratulations! Use ₹3.5L to pay off HDFC & ICICI, then enjoy one simple EMI.

Thank you for choosing Tata Capital! 🎉"

! EDGE CASE 1: Low Credit Score

Flow:

USER: "I need loan" → Provides PAN

CREDIT BUREAU AGENT:

"📊 Credit Profile:

- HDFC Loan: ₹1,50,000
- ICICI Loan: ₹1,00,000
- Credit Score: 650/900 ❌
- Minimum Required: 700"

UNDERWRITING AGENT:

"❌ Cannot Approve Consolidation

Your Financial Health Score: 5/10

- Credit Score: 650 (Needs +50 points)
- Debt-to-Income: High

💡 Improvement Plan:

1. Pay EMIs on time for 3 months (+30 points)
2. Reduce credit card usage (+20 points)
3. Reapply at 700+ score

Meanwhile, here's your Financial Health Report 📄"

! EDGE CASE 2: Single Existing Loan

Flow:

USER: Provides PAN

CREDIT BUREAU AGENT:

" Credit Profile:

- Only SBI Loan: ₹2,00,000
- Credit Score: 800/900 

MASTER AGENT:

"Great credit score! For your ₹2L loan requirement:

OPTION 1 - New Loan:

- ₹2,00,000 @ 11.5% = ₹6,580/month
- You keep SBI loan separately

OPTION 2 - Loan Transfer:

- Transfer SBI ₹2L + New ₹2L = ₹4,00,000
- Single EMI: ₹12,350 @ 11%
- Save: ₹1,150/month

Prefer consolidation or separate loan?  "

EDGE CASE 3: No Existing Loans

Flow:

USER: Provides PAN

CREDIT BUREAU AGENT:

" Credit Profile:

- No Existing Loans 
- Credit Score: 750/900 

SALES AGENT:

"Perfect profile! For ₹2L personal loan:



BEST OFFER:

- Amount: ₹2,00,000
- Interest: 10.5% (Special rate for good credit)
- Tenure: 3 years
- EMI: ₹6,500/month

Ready to proceed? 

EDGE CASE 4: Salary Slip Required

Flow:

USER: Requests ₹5L loan → Provides PAN

CREDIT BUREAU AGENT:

"Existing Loans: ₹2L | Credit Score: 780 

UNDERWRITING AGENT:

"Loan Request: ₹5,00,000

Pre-approved Limit: ₹3,00,000



Additional Verification Needed:

Please upload your salary slip for higher amount approval.

[File Upload Button]"

EDGE CASE 5: Too Many Existing Loans

Flow:

CREDIT BUREAU AGENT:

" Credit Profile:

- HDFC: ₹1L | ICICI: ₹1L | Axis: ₹1L | SBI: ₹1L
- Total Active Loans: 4 
- Credit Score: 700"

UNDERWRITING AGENT:

" ⚠️ Too Many Existing Loans

We recommend:

1. Pay off 2 smaller loans first
2. Improve debt-to-income ratio
3. Reapply for consolidation later

Your Financial Health Report shows step-by-step plan 📄 "

KEY FEATURES DEMONSTRATED:

- ✅ **Auto-detection** via credit bureau
- ✅ **Smart routing** based on loan count & score
- ✅ **Edge case handling** (low score, single loan, etc.)
- ✅ **Professional PDF output**
- ✅ **End-to-end in one chat**
- ✅ **Business logic compliance**
- ✅ **Hero feature integration**

repo structure:

```
tata_loan_chatbot/
|
|— app.py                # Main Streamlit app (Rahul)
|— master_agent.py       # Orchestrator (Aakash)
|— requirements.txt      # Dependencies
|
|— agents/
|   |— sales_agent.py    # Loan offers (Angad)
|   |— verification_agent.py # KYC & identity (Angad)
|   |— credit_bureau_agent.py # Auto loan detection (Aakash)
|   |— underwriting_agent.py # Approval logic (Aakash+Swati)
|   |— document_agent.py # PDF generation (Sanvi)
```



```
|
|— data/
|   |— customers.json          # Customer profiles (Sanvi) ✓
|   |— credit_bureau_data.json # Mock loan data (Sanvi) ✓
|
|— utils/
|— loan_calculator.py        # EMI calculations (Sanvi) ✓
```